

HIGH-PERFORMANCE & PARALLEL COMPUTING

Lab 1: The Fork-Join Model, Team Creation, and Thread Scoping

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Huawei Laptop MateBook D15

Windows 11

11th Gen Intel(R) Core(TM) i5- 1135G7 @ 2.40GHz

Physical Cores: 4

Logical SMT Threads: 8

Cache Hierarchy Capacity: L1: 320 KB | L2: 5 MB | L3: 8 MB

Task 1.1:

Ran `run_team(4)` 10 times. Logical ranks were fixed (0–3). The Master (rank 0) always finished first because its sleep time was 0 ms. Among workers, the [START] and [END] order varied each run.

Conclusion: Thread execution order is non-deterministic; it is governed by the OS scheduler and CPU core availability.

===== Task 1.1 Run 1 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 14004 | Thread Name: omp-worker-0

[END] Master | Logical Rank: 0 of 4 | Native OS TID: 14004

[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 11616 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 11068 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 4752 | Thread Name: omp-worker-3

[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 4752

[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 11616

[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 11068

===== Task 1.1 Run 2 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 30372 | Thread Name: omp-worker-0

[END] Master | Logical Rank: 0 of 4 | Native OS TID: 30372

[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 19436 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 21516 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 3036 | Thread Name: omp-worker-3

[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 3036

[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 19436

[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 21516

===== Task 1.1 Run 3 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 26368 | Thread Name: omp-worker-0

[END] Master | Logical Rank: 0 of 4 | Native OS TID: 26368

[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 14304 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 28200 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 8752 | Thread Name: omp-worker-3

[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 8752

[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 14304

[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 28200

===== Task 1.1 Run 4 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 3028 | Thread Name: omp-worker-0

[END] Master | Logical Rank: 0 of 4 | Native OS TID: 3028

[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 10260 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 14452 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 12484 | Thread Name: omp-worker-3

[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 12484

[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 10260

[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 14452

===== Task 1.1 Run 5 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 20748 | Thread Name: omp-worker-0

[END] Master | Logical Rank: 0 of 4 | Native OS TID: 20748

[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 21476 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 8812 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 9784 | Thread Name: omp-worker-3

[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 9784

[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 21476

[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 8812

===== Task 1.1 Run 6 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 26180 | Thread Name: omp-worker-0

[END] Master | Logical Rank: 0 of 4 | Native OS TID: 26180

[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 28880 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 28020 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 29124 | Thread Name: omp-worker-3

[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 29124

[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 28880

[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 28020

===== Task 1.1 Run 7 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 11948 | Thread Name: omp-worker-0

[END] Master | Logical Rank: 0 of 4 | Native OS TID: 11948

[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 23464 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 13464 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 28184 | Thread Name: omp-worker-3
[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 28184
[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 23464
[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 13464

===== Task 1.1 Run 8 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 11132 | Thread Name: omp-worker-0
[END] Master | Logical Rank: 0 of 4 | Native OS TID: 11132
[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 27320 | Thread Name: omp-worker-1
[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 3116 | Thread Name: omp-worker-2
[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 11084 | Thread Name: omp-worker-3
[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 11084
[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 27320
[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 3116

===== Task 1.1 Run 9 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 28872 | Thread Name: omp-worker-0
[END] Master | Logical Rank: 0 of 4 | Native OS TID: 28872
[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 16468 | Thread Name: omp-worker-1
[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 29240 | Thread Name: omp-worker-2
[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 11060 | Thread Name: omp-worker-3
[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 11060
[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 16468
[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 29240

===== Task 1.1 Run 10 =====

[START] Master | Logical Rank: 0 of 4 | Native OS TID: 2888 | Thread Name: omp-worker-0
[END] Master | Logical Rank: 0 of 4 | Native OS TID: 2888
[START] Worker | Logical Rank: 1 of 4 | Native OS TID: 16552 | Thread Name: omp-worker-1

[START] Worker | Logical Rank: 2 of 4 | Native OS TID: 15176 | Thread Name: omp-worker-2

[START] Worker | Logical Rank: 3 of 4 | Native OS TID: 22836 | Thread Name: omp-worker-3

[END] Worker | Logical Rank: 3 of 4 | Native OS TID: 22836

[END] Worker | Logical Rank: 1 of 4 | Native OS TID: 16552

[END] Worker | Logical Rank: 2 of 4 | Native OS TID: 15176

Task 1.2:

Measured time to create and join a thread team for $P = 1, 2, 4, 8, 16, 32, 64$. Three runs per P ; mean shown.

P	run1	run2	run3	mean
1	0.000886	0.000353	0.000248	0.000496
2	0.001774	0.002079	0.001812	0.001888
4	0.003561	0.003815	0.003352	0.003576
8	0.003171	0.003025	0.003067	0.003088
16	0.004493	0.003986	0.00408	0.004186
32	0.006542	0.006063	0.00598	0.006195
64	0.010732	0.01064	0.010147	0.010506

Observation: Time increases with P , especially after $P > 8$. This is due to oversubscription: more threads than physical cores cause context switching and scheduling overhead.

Task 1.3:

Each thread performed 10,000,000 square-root computations. Wall-clock time:

P	wall_sec
1	0.016478
2	0.031817
4	0.061993
8	0.12576
16	0.24587

Observation: Time scales linearly with P . No parallel speedup was achieved because Python's GIL prevents simultaneous execution of CPU-bound bytecode, even with Numba nogil=True. True parallelism would require multiprocessing or a GIL-free runtime.

```

import sys

import time

import threading


import numpy as np
from numba import njit


@njit
def burn_cpu(n):
    s = 0.0
    for i in range(n):
        s += np.sqrt(i)
    return s


def worker_task(thread_id, team_size, workload=0, verbose=True):
    native_tid = threading.get_native_id()
    role = "Master" if thread_id == 0 else "Worker"

    if verbose:
        print(
            f"[START] {role} | Logical Rank: {thread_id} of {team_size} | "
            f"Native OS TID: {native_tid} | Thread Name: {threading.current_thread().name}"
        )

    time.sleep(0.001 * (thread_id % 3))

    if workload > 0:
        burn_cpu(workload)

```

```
if verbose:
```

```
    print(
        f"[END] {role} | Logical Rank: {thread_id} of {team_size} | "
        f"Native OS TID: {native_tid}"
    )
```

```
def run_team(num_threads, workload=0, verbose=True):
```

```
    threads = []
```

```
    t0 = time.perf_counter()
```

```
    for tid in range(num_threads):
```

```
        th = threading.Thread(
            target=worker_task,
            args=(tid, num_threads, workload, verbose),
            name=f"omp-worker-{tid}",
        )
        threads.append(th)
        th.start()
```

```
    for th in threads:
```

```
        th.join()
```

```
    t1 = time.perf_counter()
```

```
    return t1 - t0
```

```
if __name__ == "__main__":
```

```
    burn_cpu(1000) # 预热 Numba
```

```
    mode = sys.argv[1] if len(sys.argv) > 1 else "task1"
```

```

if mode == "task1":
    for r in range(10):
        print(f"\n===== Task 1.1 Run {r + 1} =====")
        run_team(4, workload=0, verbose=True)

elif mode == "sweep":
    print("P,run1,run2,run3,mean")
    for p in [1, 2, 4, 8, 16, 32, 64]:
        times = []
        for _ in range(3):
            t = run_team(p, workload=0, verbose=False)
            times.append(t)
        mean = sum(times) / len(times)
        print(f"{p},{times[0]:.6f},{times[1]:.6f},{times[2]:.6f},{mean:.6f}")

elif mode == "cpu":
    print("P,wall_sec")
    for p in [1, 2, 4, 8, 16]:
        t = run_team(p, workload=10_000_000, verbose=False)
        print(f"{p},{t:.6f}")

else:
    print("Usage: python lab1_fork_join.py [task1|sweep|cpu]")

```